PAPER

www.rsc.org/greenchem

Organometallic reactions in ionic liquids. Alkylation of aldehydes with diethylzinc

Man Chun Law,[ab] **Kwok-Yin Wong**[b] **and Tak Hang Chan***[ab]

[a] *Department of Chemistry, McGill University, Montreal, Quebec, Canada H3A 2K6*

[b] *Department of Applied Biology and Chemical Technology, The Hong Kong Polytechnic University, Hung Hom, Hong Kong SAR, China. E-mail: bcchanth@polyu.edu.hk; Fax: +852-2364-9932; Tel: +852-2766-5605*

Received 2nd February 2004, Accepted 9th March 2004

First published as an Advance Article on the web 25th March 2004

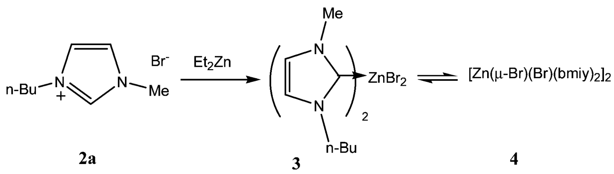
The ionic liquids, [bmim][Br] (**2a**), [bmim][BF_4] (**2b**), [bmim][PF_6] (**2c**), [bdmim][BF_4] (**8**) and [bpy][BF_4] (**9**), were examined as the solvent media for dialkylzinc addition to aldehydes giving the corresponding alcohols. The ionic liquid [bpy][BF_4] was found to be the solvent of choice, giving the best yields, easily recovered and reused.

Because of the concern for the environment, there has been considerable recent research into replacing the use of volatile organic solvents with clean solvents[1] as reaction media for chemical synthesis.[2] Water,[3] supercritical carbon dioxide[4] and ionic liquids[5] are currently the clean solvents being investigated most extensively. A common problem that is faced by these solvents is their reactivity towards reactive organometallic reagents such as the Grignard or the organolithium reagents. These organometallic reagents are very useful in organic synthesis for the formation of carbon–carbon bonds. They are well known to react with water or carbon dioxide. It is also likely that Grignard or organolithium reagents will react with quaternary salts, structures which are common for ionic liquids.[6] Alternative organometallic reactions must be sought for these solvents to replace the role of Grignard or organolithium reactions. Recently, we and others have investigated the Barbier–Grignard type reactions in aqueous media using In,[7] Zn[8] and Sn[9] and other metals as alternatives to the classical magnesium based Grignard reaction.[10] While these aqueous organometallic reactions have been found to be useful in organic synthesis, they also have limitations. Firstly, the types of reactions have been largely limited to allylation and its variations.[11] Efforts to extend them to alkylation reactions have only limited success.[12] Secondly, water sensitive substrates such as hydrolytically reactive imines cannot be used.[13] Finally, highly hydrophobic compounds insoluble in water tend to present difficulties in these reactions. We are therefore interested in examining organometallic reactions in ionic liquids and comparing their chemistry with those in aqueous media and in conventional organic solvents. The hope is that a broader range of organometallic reactions can be carried out in ionic liquids than in aqueous media. Thus far, few stoichiometric organometallic reactions in ionic liquids have been examined.[14,15] Recently we and others have found that zinc was quite effective in mediating the coupling of allyl bromide with carbonyl compounds in the ionic liquid 3-butyl-1-methylimidazolium tetrafluoroborate, [bmim][BF_4], to give the product homoallylic alcohols in good yields.[16,17] The results suggested that the presumed allylzinc intermediate, if formed under those conditions, may not have reacted with the ionic liquid solvent. This observation prompted us to examine the possibility of conducting alkylation reactions with the more reactive diethylzinc in ionic liquids.

Initially, we examined the reaction of neat diethylzinc (**1**) with [bmim][Br] (**2a**) at room temperature. Gas bubbles evolved quickly and the ionic liquid turned from pale yellow to white with increased viscosity. Based on ^1H and ^{13}C NMR together with mass spectral studies and in analogy with the known *bis*(imidazolylidene)palladium and nickel dibromide complexes,[18] the products of the reaction were assigned to have the [$ZnBr_2(bmiy)_2$] (**3**) and [Zn(μ-Br)(Br)$(bmiy)_2]_2$ (**4**) (bmiy = 3-butyl-1-methyl-2-imidazolylidene) structures (Scheme 1). Previously, imidazol-2-ylidene zinc adducts have been prepared by a different route.[19] However, when the reaction of diethylzinc with [bmim][BF_4] (**2b**) was examined, the reaction was found to be slow and the conversion to the carbene zinc complexes was incomplete. The reaction could be accelerated by the addition of NaBr. We suspected therefore that the carbene formation in [bmim][BF_4] may actually be due to the contamination of the precursor – [bmim][Br]. Indeed, when [bmim][BF_4] had been meticulously purified using the chromatography method to be free of bromide ion,[20] its reaction with diethylzinc was now sufficiently slow that it could be used as solvent for the alkylation reactions. The results are summarized in Table 1. Using [bmim][BF_4] as solvent, various aldehydes **5** were alkylated by diethylzinc to the alkylated product alcohols **6** together with the reduced alcohols **7** as the minor product (Scheme 2). In many cases where a small amount (0.05 mL) of the ionic liquid was used, a 0.5 mole ratio of Et_2Zn was able to give good yield of the adduct **6**, suggesting that both ethyl groups can be transferred to the carbonyl compound. This is distinctly different from organozinc reactions in organic solvents which have been generally described as sluggish: diorganozincs usually add to aldehydes in synthetically useful yields only in the presence of catalysts such as Lewis acids and Lewis bases.[21] For example, the reaction of diethylzinc and 4-chlorobenzaldehyde in organic solvent without catalyst gave a mixture of the adduct alcohol **6** in 38% yield together with 45% of the reduction product **7**.[21c] Aliphatic aldehydes (entry 20) were alkylated similarly in IL. For conjugated aldehydes such as *trans*-cinnamaldehyde (entry 21), alkylation occurred in a 1,2-fashion. 4-Hydroxybenzaldehyde (entry 19) was alkylated without the need to protect the hydroxyl group. However, excess diethylzinc was required to give a good yield. When [bmim][PF_6] (**2c**) was used as the ionic liquid in place of [bmim][BF_4] (**2b**), the yield of the adduct **6** was much reduced (entry 7). The major product was found to be the reduction of the aldehyde to the corresponding alcohol **7** due to hydride transfer from diethylzinc. It should be pointed out that under the solvent-free conditions,[22] reaction of aldehyde with diethylzinc did not give a satisfactory result as the reaction mixture solidified readily and gave incomplete reaction (entry 8). This is typical of the reactions of solid substrates in solvent-free conditions.

In spite of these results, the use of [bmim][BF_4] as solvent was considered unsatisfactory due to the difficulty of its purification. Chromatographic purification of [bmim][BF_4] according to literature procedures[20] was tedious and the recovery was low. The alternative method of using $AgBF_4$ to remove bromide ion was expensive[20] and the trace of silver ion remaining was found to catalyze the decomposition of diethylzinc. In addition, the reaction

DOI: 10.1039/b401520g



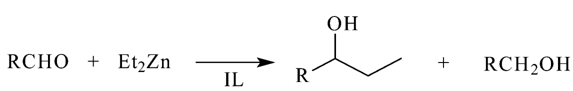
2a 3 4

Scheme 1

Table 1 Yields of alkylation reactions of various carbonyl compounds with diethylzinc in ionic liquid **2b**[a]

Entry	Carbonyl compound	Mole ratio Et_2Zn : C=O	Vol. of IL/mL	Yield **6** (%)[b]
1	PhCHO	0.5 : 1	0.05	71
2	2-FC_6H_4CHO	0.5 : 1	0.05	93
3	4-FC_6H_4CHO	0.5 : 1	0.05	68
4	2-ClC_6H_4CHO	0.5 : 1	0.05	61
5	3-ClC_6H_4CHO	0.5 : 1	0.05	99
6	2,6-$Cl_2C_6H_3CHO$	2 : 1	2.00	68
7[c]	2,6-$Cl_2C_6H_3CHO$	1 : 1	1.00	39
8[d]	2,6-$Cl_2C_6H_3CHO$	1 : 1	neat	29
9	4-EtC_6H_4CHO	0.5 : 1	0.05	40
10	2-$MeOC_6H_4CHO$	2 : 1	2.00	69
11	3-$MeOC_6H_4CHO$	0.5 : 1	0.05	38
12	4-$MeOC_6H_4CHO$	2 : 1	2.00	75
13	2-MeC_6H_4CHO	0.5 : 1	0.05	70
14	3-MeC_6H_4CHO	0.5 : 1	0.05	(59)
15	3-MeC_6H_4CHO	1 : 1	0.05	84
16	4-MeC_6H_4CHO	0.5 : 1	0.05	(30)
17	4-MeC_6H_4CHO	1 : 1	0.05	91
18	1-naphthaldehyde	0.5 : 1	0.05	50
19	4-HOC_6H_4CHO	2 : 1	1.00	78[e]
20	2-methylvaleraldehyde	0.5 : 1	0.05	94
21	*trans*-cinnamaldehyde	0.5 : 1	0.05	67
22	4-CNC_6H_4CHO	2 : 1	0.50	86

[a] Addition of diethylzinc to carbonyl compounds was carried out at 1 mmol scale of the carbonyl compound at 0° → room temperature in ionic liquid (**2b**) overnight. [b] Isolated yields (NMR yield in parenthesis). [c] [bmim][PF_6] (**2c**) was used as the ionic liquid. [d] Reaction was carried out neat without IL. The reaction stopped in 5 minutes and could not be stirred further. [e] 4-Hydroxybenzaldehyde was first dissolved in ionic liquid before addition to diethylzinc.



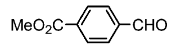
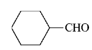
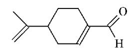
5 1 6 7

Scheme 2

often had to be carried out at 0 °C to reduce the reaction of bmim with diethylzinc. We therefore examined two other ionic liquids, 3-butyl-1,2-dimethylimidazolium tetrafluoro-borate [bdmim][BF_4] (**8**) and N-butylpyridinium tetrafluoroborate [bpy][BF_4] (**9**), for the alkylation reactions. These two solvents do not react with diethylzinc to give the carbene complexes since they do not have the relatively acidic hydrogen at the C2 position of the imidazole ring. There was no need for extensive purification to remove bromide ion from these two ionic liquids. The reactions of diethylzinc with aldehydes are summarized in Tables 2 and 3. Using 2,6-dichlorobenzaldehyde as the common substrate, 1 mL of the ionic liquid as solvent and reaction with one mole of diethylzinc at room temperature for two hours as the standard conditions, [bpy][BF_4] was the best solvent in giving essentially quantitative yield of **6**, with little of the reduction product **7** (Table 3, entry 1). Though inferior to [bpy][BF_4], [bbdmim][BF_4] appeared to be superior to [bmim][PF_6] and [bmim][BF_4] (Table 2, entry 1).

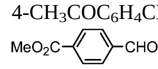
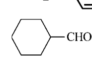
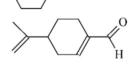
We have also studied the diethylzinc alkylation of a number of other aldehydes in these two solvents. In all cases, excellent isolated yields of the adducts could be obtained using [bpy][BF_4] as the solvent. The yields in [bdmim][BF_4], though generally inferior,

Table 2 Yields of alkylation reactions of various carbonyl compounds with diethylzinc in IL **8**[a]

Entry	Substrate (**5**)	Mole ratio **5 : 1**	Time	Yield (%)[b] **6**	**7**
1	2,6-$Cl_2C_6H_5CHO$	1 : 1	2 h	73	6
2	PhCHO[c]	1 : 0.7	2 h	80	17
3	PhCHO[c]	1 : 1	2 h	85	6
4	PhCHO	1 : 1	2 h	88	8
5	4-CNC_6H_4CHO	1 : 1	6 h	(48)	19
6	2,3-$Cl_2C_6H_5CHO$	1 : 1	5 h	(70)	0
7	4-HOC_6H_4CHO	1 : 2	10 min[d]	33	2
8	4-$CH_3COC_6H_4CHO$	1 : 1	5 min	(86)	0
9		1 : 1	5 min	(63)	0
10		1 : 1	3 h	(58)	—
11		1 : 2	12 h	(81)	4
12	PhCH=CHCHO	1 : 1	12 h	(97)	—
13	Hexanal	1 : 1	3 h	(61)	—

[a] Addition of diethylzinc to carbonyl compound was carried out at a 1 mmol scale of carbonyl compound at room temperature in ionic liquid **8** (1 mL unless noted otherwise) with vigorous stirring of the reaction mixture. [b] Measured by ^1H NMR (isolated yields in parentheses). [c] 0.1 mL of IL **8** was used. [d] The reaction mixture could not be further stirred and the reaction was stopped.

Table 3 Yields of alkylation reactions of various carbonyl compounds with diethylzinc in IL **9**[a]

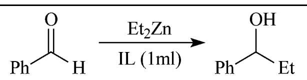
Entry	Substrate (**5**)	Mole ratio **5 : 1**	Time	Yield (%)[b] **6**	**7**
1	2,6-$Cl_2C_6H_5CHO$	1 : 1	2 h	(90)	0
2	PhCHO	1 : 0.7	2 h	(87)	8
3	4-CNC_6H_4CHO	1 : 1	6 h	(89)	8
4	2,3-$Cl_2C_6H_5CHO$	1 : 1	5 h	(88)	5
5	4-HOC_6H_4CHO	1 : 1	6 h	34	3
6	4-HOC_6H_4CHO	1 : 2	6 h	72	4
7	4-HOC_6H_4CHO	1 : 3	6 h	(76)	6
8	4-$CH_3COC_6H_4CHO$	1 : 1	5 min	(86)	0
9		1 : 1	5 min	(90)	0
10		1 : 1	3 h	(82)	12
11		1 : 2	12 h	(96)	0
12	PhCH=CHCHO	1 : 1	12 h	(92)	0
13	Hexanal	1 : 1	3 h	(75)	—

[a] Addition of diethylzinc to carbonyl compound was carried out at a 1 mmol scale of carbonyl compound at room temperature in ionic liquid **9** (1 mL) with vigorous stirring of the reaction mixture. [b] Measured by ^1H NMR (isolated yields in parentheses).

were still adequate. It is interesting to note that the ethylation reaction was chemoselective. Aryl ketone and ester functions (entries 8 and 9 in Tables 2 and 3) were not alkylated under these conditions. Aliphatic aldehydes were alkylated readily to give adducts in satisfactory yields (entries 10 and 13). Ethylation of conjugated aldehydes occurred in a 1,2-fashion (entry 12). In many cases, the side reaction, due to hydride transfer, leading to the reduced alcohols **7** appeared to have been suppressed when [bpy][BF_4] was used as the solvent.

One problem anticipated in the use of ionic liquids as solvents for organometallic reactions is the issue of recycling of the ionic liquids. In the present reaction, the inorganic zinc salt [presumably $Zn(OH)_2$] after aqueous work-up would remain in the recovered ionic liquid and may interfere with subsequent alkylation reactions. Indeed, when [bpy][BF_4] was recovered, dried vigorously under vacuum at 100 °C and reused as solvent for the ethylation of benzaldehyde, the yield of the product alcohol **6** was reduced to

Table 4 Recycling of [bpy][BF_4][a]



IL/No. of cycle	Yield[b] 1	2	3	4
Drying alone	68	50		
CH_2Cl_2 treatment	92	89	88	85
Oxalic acid treatment	93	86		

[a] Addition of diethylzinc to carbonyl compound was carried out at a 1 mmol scale of each reagent at room temperature in ionic liquid **9** (1 mL) with vigorous stirring. [b] Isolated yield.

68% (Table 4). The yield was further reduced to 50% on the next cycle. We found, however, if the inorganic zinc salt was removed from the recovered [bpy][BF_4] by addition of a small amount of methylene chloride, the ionic liquid can be reused readily to give the alkylation product with qualitatively the same rates and similar yields for several cycles (Table 4). Alternately, the reaction mixture could be quenched with oxalic acid instead of water, and the resultant zinc oxalate removed from the recovered ionic liquid by the addition of a small amount of acetone. The [bpy][BF_4] thus recovered could also be reused for the reaction to give the ethylated product in similar yields. There was little loss of the ionic liquid in the recycling process since the extract in each work-up was free of ionic liquid according to its NMR.

In conclusion, we have found that alkylation of aldehydes with diethylzinc can be conducted in ionic liquids. Using N-butylpyridinium tetrafluoroborate as the solvent, the reaction proceeded cleanly with high yield. This is in contrast to the reaction in aqueous media where diethylzinc is hydrolyzed readily, or in conventional volatile organic solvents where the addition of diethylzinc to aldehyde is generally sluggish and gives a low yield.[21] Finally, it was recently reported that [bmim][$F{\cdot}H_2O$] was identified as a decomposition product during purification of [bmim][PF_6].[23] Since HF is considered toxic, it has been cautioned that ionic liquids are not always green.[23] Even though at this time, we do not have evidence to suggest that either [bdmim][BF_4], or [bpy][BF_4] decompose to give HF containing products, caution must be exercised as with any ionic liquids with no supporting toxicological data.

Experimental

Neat diethylzinc was purchased from Aldrich and was used without further purification. All manipulations were carried out using Schlenk techniques under inert nitrogen atmosphere. The liquid diethylzinc was kept under nitrogen and transferred with a syringe into the reaction mixture. Nitrogen gas must flow into the container by piercing through the septum fitted on top with a needle during withdrawal of diethylzinc. Excess diethylzinc was quenched with acetone. Diethylzinc should be handled with care and not be exposed to aqueous media as this will lead to a fire hazard.[24]

Preparation of ionic liquids

Ionic liquids [bmim][PF_6], [bmim][BF_4], [bdmim][BF_4] and [bpy][BF_4] were prepared by metathesis with sodium tetrafluoroborate from its bromide precursors which were in turn prepared from the microwave-assisted method.[25] The ionic liquids were vacuum-dried overnight at 70 °C (0.1 mmHg) and stored under argon.

General procedure for alkylation with diethylzinc in ionic liquid

A 30 mL three-necked round-bottomed flask was charged with the ionic liquid (0.1 to 1 mL, as specified in Tables 1 and 2). Diethylzinc (0.5 to 3 mmol) was added dropwise *via* syringe to the ionic liquid at room temperature under nitrogen. The aldehyde (1 mmol) was then added dropwise under nitrogen flow throughout the addition process and the reaction mixture was stirred vigorously at room temperature for the specified time. The resulting mixture was quenched with a few drops of water followed by extraction with Et_2O (2×5 mL).† After removal of the ether solvent *in vacuo*, the residue was purified by flash chromatography (10 : 1 hexane : EtOAc as eluent) on silica gel to yield the pure adduct alcohol **6**. Specific conditions and yields of the alkylation of aldehydes are given in Tables 1, 2 and 3.

Recycling and reuse of ionic liquids

After the reaction was completed and work-up carried out as described above, the ionic liquid could be reused after drying at 100 °C (0.1 mmHg) without further treatment. However this would result in the accumulation of inorganic zinc salt residues and reduce the yield of subsequent reactions. Method A: the zinc residue was removed by dissolution of the ionic liquid (1 mL) in methylene chloride (4 mL) and the solid precipitate of $Zn(OH)_2$ was filtered. The ionic liquid was recovered on removal of methylene chloride and dried by further heating overnight at 70 °C (0.1 mmHg). The experimental procedure for alkylation with diethylzinc in recycled ionic liquid was the same as described above. Method B: after complete reaction, 1 equiv. of oxalic acid dissolved in acetone (1 mL) was added to quench the reaction and stirred for 1 hour. More acetone (3 mL) was added to the mixture and the solid precipitate of ZnC_2O_4 was filtered. The filtrate was subjected to rotary evaporation to remove the acetone, followed by extraction with diethyl ether. The ionic liquid, recovered on removal of the diethyl ether, was dried by further heating at 70 °C (0.1 mmHg). The experimental procedure for alkylation with diethylzinc in the recycled ionic liquid was the same as described above except the reaction mixture was heated at 40 °C after stirring overnight for a few hours.

3-Butyl-1-methylimidazolylidene zinc complexes

A three-necked 50 ml round bottom flask was charged with 1-methyl-3-butyl-1-methylimlidazolium bromide (0.5 g, 2.28 mmol) followed by dropwise addition of diethylzinc (0.11 mL, 1.14 mmol). The flask was capped with a septum, and the rate of gas evolution was monitored by a cranial tube connected to an oil-filled bubbler. After completion of the reaction as indicated by the cessation of gas evolution in 5 mins, the mixture was examined by NMR. The product was found to be a mixture of **3** and **4** in a ratio of about 1 : 4.

[$ZnBr_2(bmiy)_2$] (3)

^{1}H NMR (CD_3CN): 7.46 (s, 2H), 7.43 (s, 2H,), 4.16 (t, 4H), 3.77 (s, 6H), 1.70 (m, 4H), 1.30 (m, 4H), 0.89 (t, 6H). ^{13}C NMR (CD_3CN): 173.9, 124.0, 122.3, 50.3, 37.7, 33.7, 20.0, 13.6.

[$Zn(\mu\text{-}Br)(Br)(bmiy)_2]_2$ (4)

^{1}H NMR (CD_3CN): 7.34 (s, 2H), 7.31 (s, 2H), 4.37 (t, 4H), 3.96 (s, 6H), 1.77 (m, 4H), 1.35 (m, 4H), 0.94 (t, 6H); ^{13}C NMR (CD_3CN): 173.6, 123.4, 121.7, 50.1, 37.7, 34.0, 19.9, 13.7.

MS (FAB): *m/z* 139 (100); 203 (18), cluster; 281 (2.7), cluster; 419 (1.6), cluster; 289 (2.1), 303 (4.3), 317 (9.0), 331 (11.4), 345 (8.5), 359 (4.3) and 373 (1.4) clusters; 413 (0.8), 427 (1.0), 441 (1.7), 455 (3.2), 469 (4.1), 483 (2.8), 497 (1.3) and 511 (0.4) clusters; 760 (0.4), 774 (0.9), 788 (1.5), 802 (2.2), 816 (2.6), 830 (2.5), 844 (1.7), 858 (0.8), 872 (0.3) and 886 (0.1) clusters.

† It is assumed generally that, if necessary, extraction of organic substrates from ionic liquids can be carried out with supercritical carbon dioxide in place of non-polar organic solvents. See ref. 26.

Acknowledgements

We thank NSERC and HKRGC for financial support of this research.

References

1 *Clean Solvents: Alternative Media for Chemical Reactions and Processing*, ed. M. Abraham and L. Moens, ACS Symp. Ser. No. 819, American Chemical Society, Washington, DC, 2001.
2 P. T. Anastas and J. C. Warner, *Green Chemistry: Theory and Practice*, Oxford Science Publications, Oxford, 1998.
3 C. J. Li and T. H. Chan, *Organic Reactions in Aqueous Media*, John Wiley & Sons, New York, 1997.
4 *Chemical Synthesis Using Supercritical Fluids*, ed. P. Jessop and W. Leitner, Wiley-VCH, Weinheim, 1999.
5 J. D. Holbrey and K. R. Seddon, *J. Chem. Soc., Dalton Trans.*, 1999, 2701; J. S. Wilkes, *Green Chem.*, 2002, **4**, 73.
6 For recent examples of reactions of ionic liquids with reactive organometallic reagents, see D. S. McGuinness, W. Mueller, P. Wasserscheid, K. J. Cavell, B. W. Skelton, A. H. White and U. Englert, *Organometallics*, 2002, **21**, 175.
7 C. J. Li and T. H. Chan, *Tetrahedron Lett.*, 1991, **32**, 7017; T. H. Chan and Y. Yang, *J. Am. Chem. Soc.*, 1999, **121**, 3228.
8 C. Petrier and J. L. Luche, *J. Org. Chem.*, 1983, **50**, 910.
9 T. H. Chan, Y. Yang and C. J. Li, *J. Org. Chem.*, 1999, **64**, 4452.
10 T. H. Chan, C. J. Li, M. C. Lee and Z. Y. Wei, *Can. J. Chem.*, 1994, **72**, 1181.
11 C. J. Li and T. H. Chan, *Tetrahedron*, 1999, **55**, 11149.
12 C. C. K. Keh, C. Wei and C. J. Li, *J. Am. Chem. Soc.*, 2003, **125**, 4062.
13 T. Vilaivan, C. Winotapan, T. Shinada and Y. Ohfune, *Tetrahedron Lett.*, 2001, **42**, 9073.
14 T. Kitazume and K. Kasai, *Green Chem.*, 2001, **3**, 30.
15 C. M. Gordon and A. Mccluskey, *Chem. Commun.*, 1999, 1431.
16 M. C. Law, K. Y. Wong and T. H. Chan, *Green Chem.*, 2002, **4**, 161.
17 C. M. Gordon and C. Ritchie, *Green Chem.*, 2002, **4**, 124.
18 (*a*) C. J. Mathews, P. J. Smith, T. Welton, A. J. P. White and D. J. Williams, *Organometallics*, 2001, **20**, 3848; (*b*) L. Xu, W. Chen and J. Xiao, *Organometallics*, 2000, **19**, 1123.
19 A. J. Arduengo III, H. V. Rasika Dias, F. Davidson and R. L. Harlow, *J. Organomet. Chem.*, 1993, **462**, 13.
20 S. Park and R. J. Kazlauskas, *J. Org. Chem.*, 2001, **66**, 8395.
21 (*a*) P. Knochel and R. D. Singer, *Chem. Rev.*, 1993, **93**, 2117; (*b*) K. Soai and S. Niwa, *Chem. Rev.*, 1992, **92**, 833; (*c*) B. Marx, *C. R. Hebd. Seances Acad. Sci., Ser. C*, 1968, **266**, 1646.
22 K. Tanaka and T. Toda, *Chem. Rev.*, 2000, **100**, 1025.
23 R. P. Swatloski, J. D. Holbrey and R. D. Rogers, *Green Chem.*, 2003, **5**, 361.
24 For the preparation and safe handling of dialkylzinc, see; *Synthetic Methods of Organometallic and Inorganic Chemistry*, ed. W. A. Herrmann, vol. 5, Thieme, Stuttgart, 1999.
25 M. C. Law, K. Y. Wong and T. H. Chan, *Green Chem.*, 2002, **4**, 328.
26 L. A. Blanchard, D. Hancu, E. J. Beckman and J. F. Brennecke, *Nature*, 1999, **399**, 28.